Power Trade

Margin Methodology

Power Trade margin is denominated in USD. For the purposes of accounting, the exchange assumes a constant USDC/USD rate of 1.0.

The Margin engine applies SPAN methodology to the entire portfolio, with PV measured in USD terms.

Methodology

Margin requirement is computed periodically using real-time data. At 100ms intervals, the margin engine samples spot, futures, perpetuals and options mark prices.

Funding costs are implied from futures and spot marks.

An implied volatility surface is generated for each pair where options are supported, using observed options marks, spot marks and implied funding costs.

Each currency pair is considered in turn. For each pair, a theoretical PV is computed for all positions that have this pair as the underlying currency. This includes spot, futures, perpetuals and options. The PV is computed in terms of USD.

For each currency pair, the inventory of non-usd coins in the customer's wallet is treated as USD converted at the current mark price plus a long synthetic overnight future. In this way, non-usd currencies are considered as margin cover in USD while providing the full benefit of covered call positions.

The PV of each currency-pair set is computed for each of nine scenarios, where inputs are the Spot Shock Factor (SSF) and Volatility Shock Factor (VSF). The Implied Volatility (IV) is interpolated for each product from the computed volatility surface using the tenor of the option and the Spot Rate Input (SRI).

It is important to note that the treatment of non-usd coins is only considered if there are held positions in the customer's account that are derivatives of the given underlying non-usd coin. In other words simply holding a non-usd coin (like ETH) does not result in an automatic haircut being applied to the value of the coin, and any treatment is capped by the impact of exposure resulting from those held positions.

For example, if the value of a held position is significantly smaller than the value held of the underlying, then a scenario which sees a value at risk of 15% of the held position which is itself only 4% of the value of the underlying, means that the scenario analysis will cap the “impact” of holding the underlying to that 4% and not assume a value at risk of 15% as expected for the scenario in play.

Definitions

SSF - Spot Shock Factor
VSF - Volatility Shock Factor
IV - Implied Volatility for a given tenor and moneyness
SRI - Spot Rate Input

Scenarios

Scenario Number	**Spot Rate Input**	**Implied Volatility Input**
1	Spot Mark	IV
2	Spot Mark	IV + (IV * VSF)
3	Spot Mark	IV - (IV * VSF)
4	Spot Mark * (1 + SSF)	IV
5	Spot Mark * (1 + SSF)	IV + (IV * VSF)
6	Spot Mark * (1 + SSF)	IV - (IV * VSF)
7	Spot Mark * (1 - SSF)	IV
8	Spot Mark * (1 - SSF)	IV + (IV * VSF)
9	Spot Mark * (1 - SSF)	IV - (IV * VSF)

Note that both the volatility shock and the spot shock are additive.

Once the nine PVs are computed, the scenario with the lowest PV is selected.

This process is repeated for each currency pair and the resulting PV of each selected scenario is accumulated.

The margin requirement is the difference between the accumulated scenario PV and the accumulated PV of the un-shocked portfolio.

Margin requirement is always stated in USD.

The margin system has the capability to set a correlation between currency pairs. At the present time, this capability is not enabled.

Configurable Parameters

The input parameters are:

- Spot Shock Factor (SSF)
- Volatility Shock Factor (VSF)

Parameters may be modified from time to time in response to adverse market conditions. However, the baseline parameters are as follows:

	ETH	**BTC**	**SOL**	**AVAX**
SSF	15%	15%	30%	30%
VSF	40%	40%	80%	80%

PowerTrade offers a Testnet where you could enter options and outright positions to see the margin calculations for a portfolio. https://powertrade-web-test.web.app

Main Account / Sub accounts Margin

The PowerTrade Margin engine calculates the margin required for each (sub)account.

To benefit from margin offset, futures and options contracts should be traded in the same (sub)account.

If two different strategies are traded via two sub-accounts, each sub-account is margined separately.